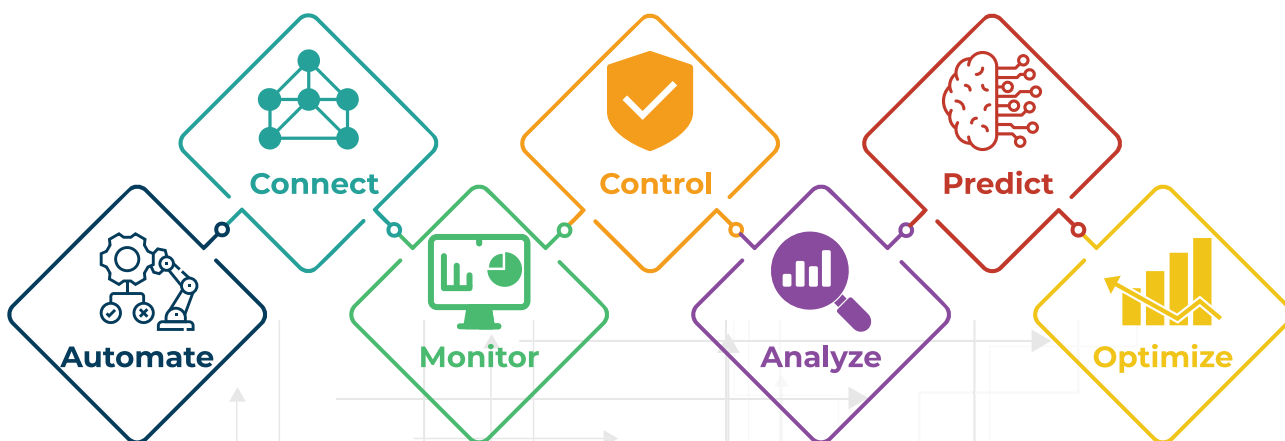




THETAVEGA

ENGINEERING THE FUTURE OF MANUFACTURING

AUTOMATION | DIGITALIZATION | TRACEABILITY



Factory Automation & Manufacturing IT

AI-Driven MES • Digital Manufacturing • Industrial Intelligence



About

Factory Automation & Manufacturing IT Solutions Partner

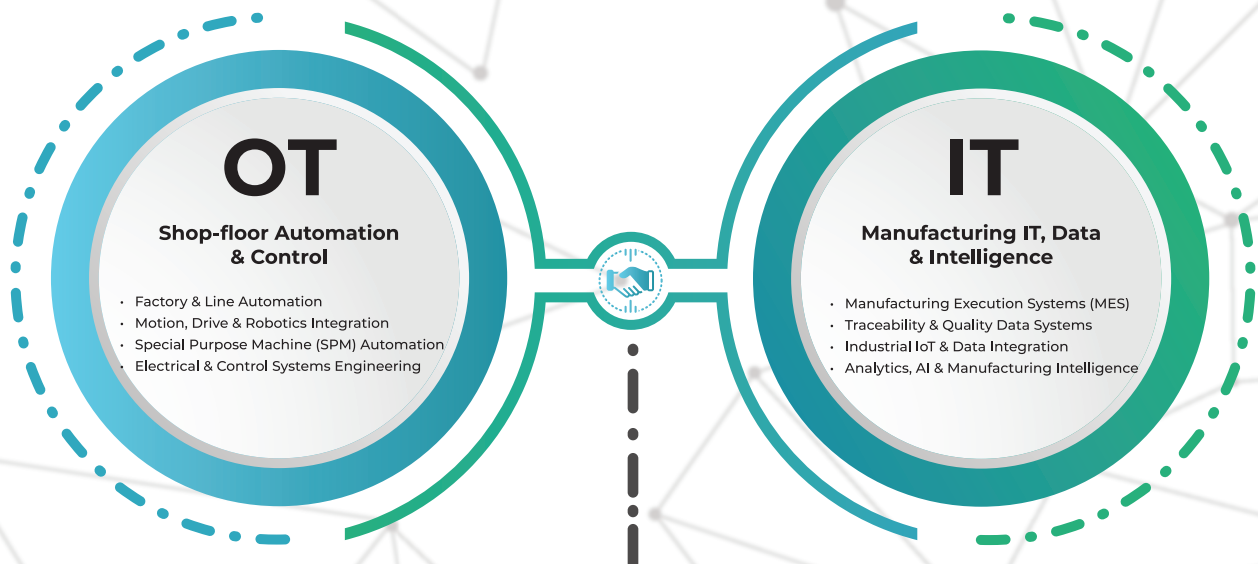
ThetaVega Tech Private Limited is a **Factory Automation (FA) and Information Technology (IT)** company delivering end-to-end shop-floor-to-enterprise solutions for manufacturing industries.

We combine automation engineering, industrial software, and manufacturing IT to enable customers to achieve controlled production, assured quality, complete traceability, and data-driven decision-making.

ThetaVega is built by engineers with deep expertise in factory automation systems and manufacturing IT platforms. Our core strength lies in OT-IT convergence, enabling seamless integration across machines, operators, production systems, and enterprise applications.

We follow a structured, compliance-oriented engineering approach covering system design, development, validation, deployment, and full lifecycle support, ensuring long-term reliability, scalability, and regulatory readiness.

**Your Partner
for End-to-End
Manufacturing
Excellence**

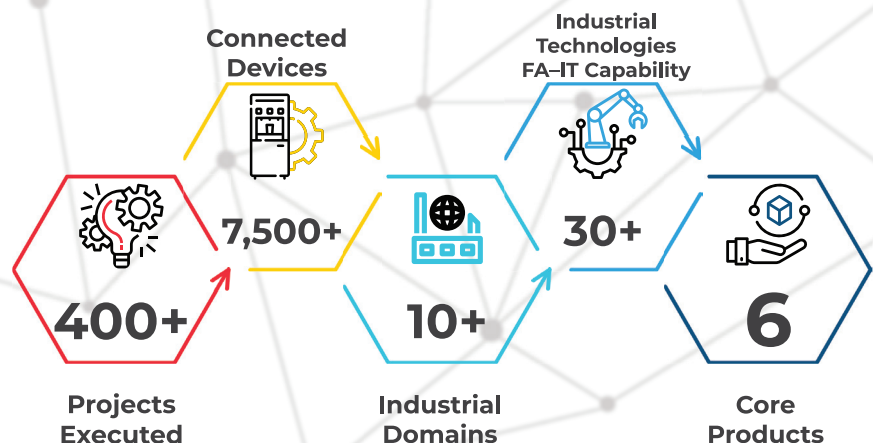


Built on International Standards

Our solutions are designed and engineered in alignment with key international standards and guidelines to ensure long-term reliability, safety, and compliance, including:

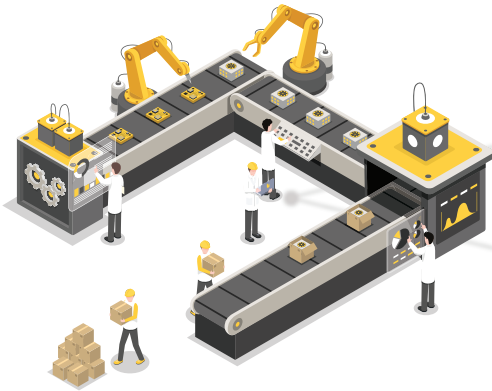


**Proven Scale &
Engineering
Depth**



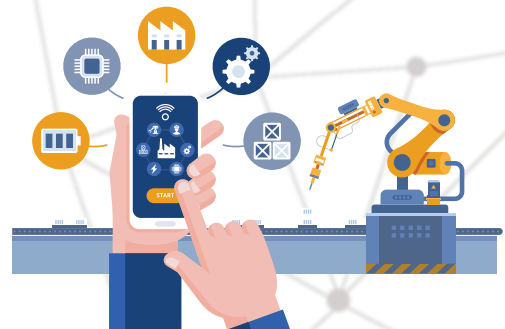
Our Core Capabilities

End-to-End Factory Automation & Manufacturing IT Expertise



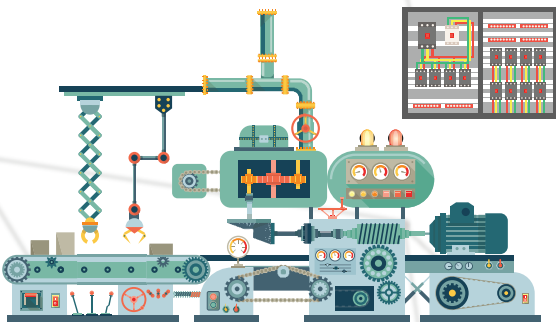
Factory Automation & Control Systems

We design and deploy PLC-based automation, robotics integration, and machine-level control solutions.



OperateX™ - Manufacturing Execution System (MES)

Our in-house platform delivers traceability, quality assurance, and production monitoring.

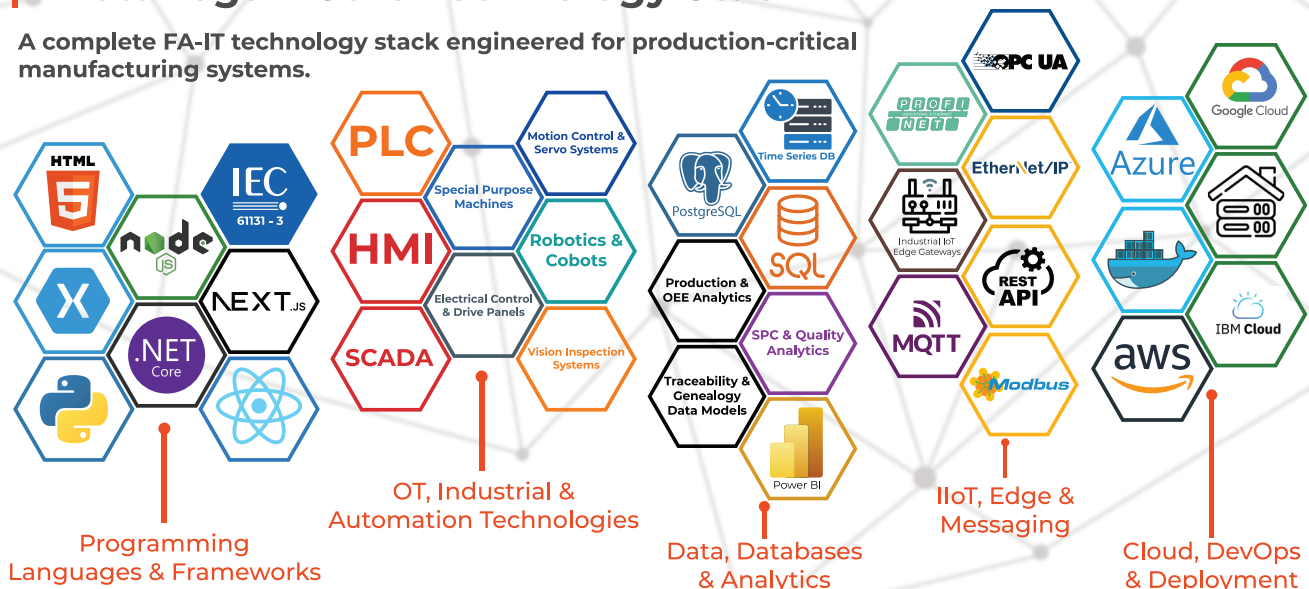


Turnkey Machine Building & Panel Building

We design and build custom machines, assembly lines, and compliant PLC/drive panels.

ThetaVega – Core Technology Stack

A complete FA-IT technology stack engineered for production-critical manufacturing systems.



Revolutionizing Manufacturing Operations

OperateX™ is a next-generation Manufacturing Execution System (MES) designed for discrete, process, and hybrid manufacturing environments. Built with deep OT-IT convergence, OperateX seamlessly connects machines, people, and enterprise systems—transforming raw shop-floor data into real-time operational intelligence.

From production planning to dispatch, OperateX ensures complete visibility, control, and traceability across your factory—enabling faster decisions, higher quality, and sustained profitability.

Business Challenges

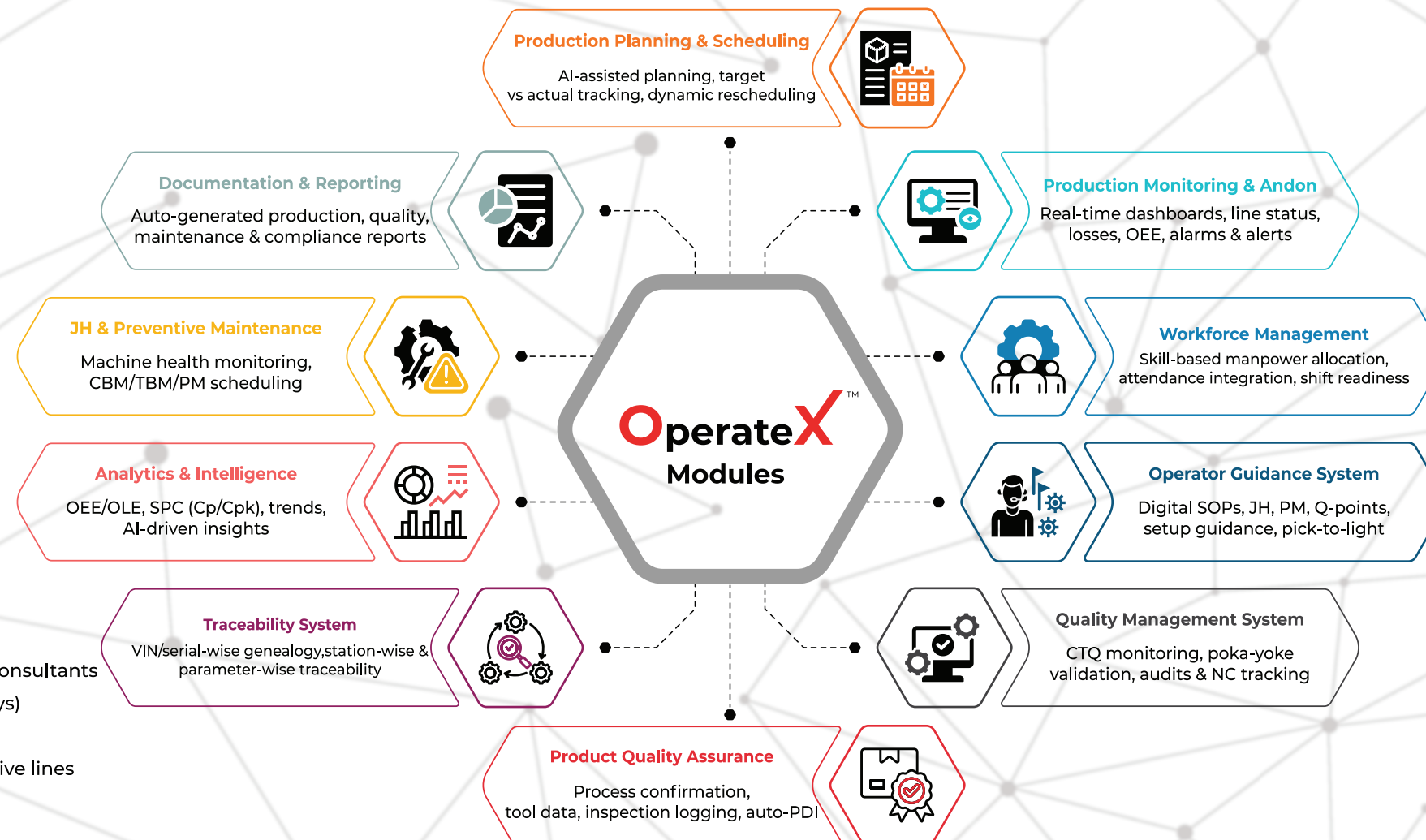
- Fragmented shop-floor data
- Manual reporting & delayed decisions
- Quality escapes & rework
- High downtime & reactive maintenance

OperateX Value Proposition

- Single Source of Truth for Manufacturing
- Real-time, event-driven decision making
- End-to-end visibility from Plan → Dispatch
- Designed for Automotive & Discrete Manufacturing Environments

OperateX Differentiators

- Built automation-first, not IT-only MES
- Native PLC, CNC, Tool & Vision integration
- Designed by factory-floor engineers, not consultants
- Event-driven architecture (no polling delays)
- Modular – start small, scale plant-wide
- Proven in high-mix, high-volume automotive lines



OperateX Architecture

L4 – ERP / SAP

Orders, BOM, Inventory, Dispatch

L3 – OperateX MES

Planning, Quality, Traceability, Analytics

L2 – SCADA / Andon / HMI

Monitoring, Alerts, Visualization

L1 – PLCs / CNC / Controllers

FX5U, iQ-R, Fanuc, Siemens, etc.

L0 – Sensors & Tools (Field Level Devices)

DC tools, Leak testers, Vision, Gauges

OperateX Technology & Integration Stack

OT Connectivity

PLCs, CNCs, Robots, Vision, Test Benches

Protocols

OPC UA, Modbus TCP, MQTT, Ethernet/IP

OperateX Supports 15+ Industrial

Protocols Directly

Data Layer

Real-time DB, Historian, Traceability DB

Applications

Web, Mobile, Smart Displays, Andon

Enterprise Integration

SAP, ERP, BI, Cloud

P

Productivity

- Real-time Production Visibility**
Live target vs actual monitoring across lines, machines, and shifts.
- OEE-Driven Performance**
Automated OEE, loss analysis, and bottleneck identification.
- Optimized Workforce & Resource Utilization**
Skill-based manpower allocation and stage-wise readiness.
- Reduced Idle & Setup Losses**
Digital SOPs, operator guidance, and guided changeover

Q

Quality

- Built-in Quality at Source**
CTQ, poka-yoke, and process confirmation enforced at every stage.
- End-to-End Product Traceability**
VIN/serial-wise genealogy with machine & process data.
- Zero Defect Dispatch Assurance**
Quality gates and system-based PDI validation before dispatch.
- Real-time Defect Detection & Alerts**
Instant visibility of NCs, deviations, and audit failures.

C

Cost

- Reduced Scrap & Rework Costs**
Early defect detection and process enforcement.
- Lower Downtime & Maintenance Costs**
Condition-based & predictive maintenance using live machine data.
- Optimized Inventory & WIP**
Real-time production, consumption, and e-Kanban visibility.
- Elimination of Manual Reporting Effort**
Automated reports for production, quality, maintenance, and audits.

D

Delivery

- Improved Schedule Adherence**
Real-time tracking of plan vs execution.
- Faster Issue Escalation & Resolution**
Andon alerts and role-based notifications.
- Reliable Dispatch Planning**
System-validated production and quality confirmation.
- Shorter Lead Time & Time-to-Market**
Streamlined workflows and reduced decision latency.

S

Safety

- Centralized Safety Monitoring**
Live status of safety devices, interlocks, and E-Stops.
- Digital Safety Audits & Compliance**
Scheduled audits with auto alerts and NC tracking.
- Operator Awareness & Guidance**
Safety instructions displayed at machine/stage level.
- Proactive Risk Identification**
Early alerts for unsafe conditions and device failures.

M

Morale

- Reduced Operator Dependency & Stress**
Guided operations through digital SOPs and visual instructions.
- Clear Role-Based Visibility**
Operators, supervisors, and managers see relevant real-time data.
- Faster Problem Resolution**
Less firefighting, more structured response.
- Data-Driven Performance Recognition**
Transparent performance metrics improve accountability and motivation.

Solutions We Offer

Integrated Automation Systems for Manufacturing Plants

ThetaVega designs and deploys automation solutions for **assembly, testing, inspection, material handling, and quality enforcement** across production-critical manufacturing operations.

- Assembly Line & Workstation Automation
- Process Interlocking & Poka-Yoke Systems
- Testing, Validation & End-of-Line (EOL) Automation
- Vision-Based Inspection & Verification Systems
- Leak Testing & Functional Test Automation
- Robotic Dispensing, Handling & Pick-and-Place Systems
- Material Handling & Conveyor Automation
- Traceability-Ready Automation Architectures

Engineering-Driven Production-Ready

Key Automation Solutions

End-to-End Automation Solutions

IoT-Ready & Data-Enabled

ThetaVega specializes in custom machine design and build, delivering SPMs engineered for **high-volume production, tight tolerances, and long-term reliability.**

- **SPM Portfolio**
 - Assembly & Joining Machines
 - Servo Press, Hydraulic Press & Press-Fit Machines
 - Screw Tightening & Torque Validation Machines
 - Leak Test Machines
 - Laser Marking & Verification Machines
 - Vision Inspection & Gauging Machines
 - Functional Testing & EOL Test Machines
 - Packaging, Sorting & Handling Machines

Machine Building Solutions

Special Purpose Machines (SPMs) & Turnkey Automation

EV Manufacturing Automation Solutions

EV Motor Assembly Lines

Modular lines with servo pressing, screw tightening, encoder fitment, and process validation

Battery Module & Pack Assembly Lines

Automated assembly integrating cell sorting, welding, clamping, and BMS installation

Battery Cell Sorting Machines

Voltage & Internal Resistance (IR) based classification

Dry Air Leak Testing Systems

Pressure decay-based leak testing

End-of-Line (EOL) Testing Systems

EV motor testing (Back EMF, vibration, insulation, performance)

Battery pack functional testing (voltage, current, insulation, relay validation)

Robotic & Vision-Based EV Automation

SCARA robot sealant dispensing systems

AI-based vision inspection for battery welding and cell polarity

Battery Welding Systems

Spot welding and laser welding systems with process monitoring

Scalable, Compliant & Future-Ready

Solutions for EV Manufacturing

Automation Systems for EV Motors, Batteries & Powertrain

Panel Building & Control Systems

Engineered Control Panels for Industrial Automation

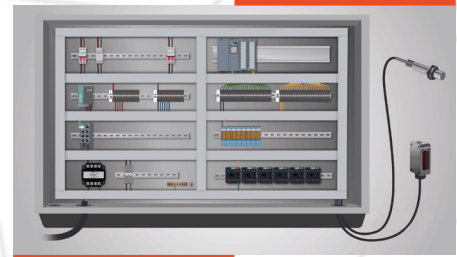
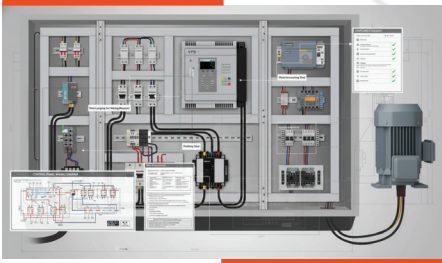
Reliable, Standards-Aligned Electrical Engineering

ThetaVega designs and builds industrial electrical control panels engineered for reliability, safety, and long-term operation in production-critical manufacturing environments.

Our panel solutions are developed with a strong focus on structured engineering, compliance, and maintainability, ensuring dependable performance across machines, lines, and plant utilities.

PLC Control Panels

- Centralized control panels for machines and production lines
- PLC-based architectures with remote I/O integration
- Designed for clear diagnostics and easy troubleshooting



Drive & Motion Control Panels

- VFD and servo drive panels for precision motion control
- Multi-axis drive systems for presses, conveyors, and robots
- Energy-efficient and performance-optimized designs

Engineering & Compliance Focus

All panels are engineered in alignment with key international standards and best practices to ensure safety, quality, and compliance

Industries We Serve

Delivering automation solutions tailored to the unique demands of each industry.



Automotive & Auto Ancillaries

Automation, machine building, testing, and traceability solutions for OEMs and Tier-1/Tier-2 suppliers.



Building Management & Smart Infrastructure

BMS, EMS, utility monitoring, and control systems for commercial, industrial, and smart infrastructure projects.



Home Appliances & Consumer Durables

Assembly automation, testing, vision inspection, and production monitoring for high-volume manufacturing



Pharmaceuticals & Life Sciences

Automation systems for regulated manufacturing environments with a focus on quality, traceability, and compliance.



Chemical & Process Industries

Control systems, batch automation, utility management, and process safety solutions for continuous and batch processes.

Food & Beverage

Automation for processing, filling, packaging, inspection, and utility management in hygiene-critical environments.



Agriculture & Agro-Processing

Automation solutions for processing plants, packaging, material handling, and utility control.



Steel, Metals & Mining

Heavy-duty automation, testing systems, utility control, and monitoring solutions for harsh industrial environments.



Textile & Garment Manufacturing

Automation and control systems for production lines, quality inspection, and plant utilities.



New Energy, Power & Gas

Automation and control systems for renewable energy, power distribution, gas handling, and energy management applications.

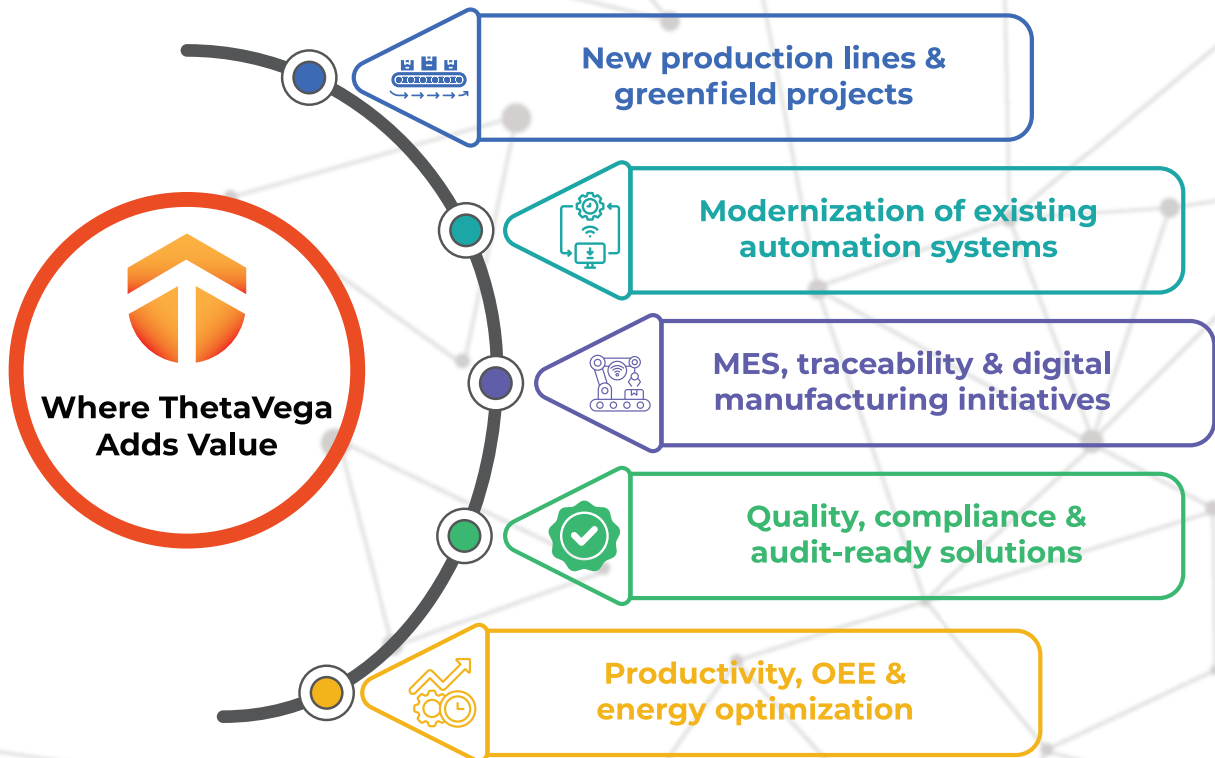


Our automation solutions are engineered to meet the unique demands of each industry.

Partner With ThetaVega

Your FA-IT Partner for Intelligent Manufacturing

ThetaVega partners with manufacturers to design, engineer, and deploy intelligent factory solutions that deliver controlled production, assured quality, complete traceability, and data-driven decision-making. By combining Factory Automation, Manufacturing IT, and AI-enabled intelligence, ThetaVega helps organizations move from isolated automation to connected, optimized, and future-ready manufacturing systems.



Let's Engineer the Digital Factory Together

Visit us to experience OperateX, our AI-driven MES, and discuss how ThetaVega can support your automation, MES, and manufacturing IT roadmap.

Get in Touch with us:



> track n trace



ThetaVega Tech Private Limited

Factory Automation & Manufacturing IT

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